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PROJECT ANALYSIS

Our application program is specially designed for the public library named

 **“READING COMMUNITY.”**

They lend books to readers who have subscribed with the library.

We have tried to maximise the efficiency and strived for customer and user ease as well as satisfaction.

We have thoroughly examined the needs of the library and after the analysis, we have constructed the program.

We have used PYTHON and MYSQL as our platform to carry out this task.

FUNCTIONS AND MODULES

Modules:

```
import mysql.connector:
```

By importing this package, we are able to establish the connection between SQL and Python.

FUNCTIONS:

connect():

This function establishes connection between Python and MySQL

cursor():

It is a special control structure that facilitates the row-by-row processing of records in the result set.

The Syntax is:

<cursor object>=<connection object>.cursor()

execute():

This function is used to execute the sql query and retrieve records using python.

The syntax is:

<cursor object>.execute(<sql query string>)

def():

A function is a block of code which only runs when it is called.

fetchall():

This function will return all the rows from the result set in the form of a tuple containing the records.

fetchone():

This Function will return one row from the result set in the form of a tuple containing the records.

commit():

This function provides changes in the database physically.

DETAILED DESCRIPTION

❖) Our Project has 3 MySQL tables. These are: -

- 1.) Books
- 2.) Issue
- 3.) Return

1) The table Books contain the following columns:

- a) bname
- b) author
- c) bcode
- d) total
- e) subject

2.) The table Issue contain the following columns:

- a.) name
- b.) regno
- c.) bcode
- d.) issue_date

3.) The table Return contain the following columns:

- a.) name
- b.) regno
- c.) bcode
- d.) return_date

SOURCE CODE

For MySQL:

```
create database library_app;  
use library_app;
```

```
create table books  
(bname varchar(50),  
author varchar(50),  
bcode varchar(50),  
total int(50),  
subject varchar(50));
```

```
create table issue  
(name varchar(50),  
regno varchar(50),  
bcode int(50),  
issue_date varchar(50));
```

```
create table return  
(name varchar(50),  
regno varchar(50),  
bcode int(50),  
return_date varchar(50));
```

For Python:

```
import mysql.connector as a
con=a.connect(host='localhost',user='root',passwd='9586',data
base='library_app')
```

def addbook():

```
    bn=input("Enter Book Name: ")
    ba=input("Enter Author's Name: ")
    c=int(input("Enter Book Code: "))
    t=int(input("Total Books: "))
    s=input("Enter Subject: ")
    data=(bn,ba,c,t,s)
    sql='insert into books values(%s,%s,%s,%s,%s);'
    c=con.cursor()
    c.execute(sql,data)
    con.commit()
    print("\n\n\n\nBook Added Successfully.....\n\n\n\n")
    wait = input('\n\n\n\nPress enter to continue.....\n\n\n\n\n\n\n')
    main()
```

def issueb():

```
    n=input("Enter Student Name: ")
    r=int(input("Enter Reg No.: "))
    co=int(input("Enter Book Code: "))
    d=input("Enter Date: ")
    a="insert into issue values(%s,%s,%s,%s);"
    data=(n,r,co,d)
    c=con.cursor()
    c.execute(a,data)
    con.commit()
    print("\n\n\n\nBook issued successfully to: ",n)
    wait = input('\n\n\n\nPress enter to continue.....\n\n\n\n\n\n\n')
    bookup(co,-1)
    main()
```

def returnb():

```
n=input("Enter Student Name: ")
r=int(input("Enter Reg No.: "))
co=int(input("Enter Book Code: "))
d=input("Enter Date: ")
a="insert into return_ values(%s,%s,%s,%s);"
data=(n,r,co,d)
c=con.cursor()
c.execute(a,data)
con.commit()
print("Book returned by: ",n)
wait = input('\n\n\nPress enter to continue.....\n\n\n\n\n\n\n')
bookup(co,1)
main()
```

def bookup(co,u):

```
a="select total from books where bcode=%s;"
data=(co,)
c=con.cursor()
c.execute(a,data)
myresult=c.fetchone()
t=myresult[0]+u
sql="update books set total=%s where bcode=%s;"
d=(t,co)
c.execute(sql,d)
con.commit()
wait = input('\n\n\nPress enter to continue.....\n\n\n\n\n\n\n')
main()
```

def dbook():

```
ac=int(input("Enter Book Code: "))
a="delete from books where bcode=%s;"
data=(ac,)
c=con.cursor()
c.execute(a,data)
con.commit()
print("Book deleted successfully")
wait = input('\n\n\nPress enter to
continue.....\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n')
main()
```

```
def dispbook():
    a="select * from books;"
    c=con.cursor()
    c.execute(a)
    myresult=c.fetchall()
    for i in myresult:
        print("Book name: ",i[0])
        print("Author: ",i[1])
        print("Book code: ",i[2])
        print("Total:",i[3])
        print("Subject:",i[4])
        print('\n\n')

    wait = input('\n\n\nPress enter to
continue.....\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n')
    main()
```

```
def report_issued_books():
    a="select * from issue;"
    c=con.cursor()
    c.execute(a)
    myresult=c.fetchall()
    for i in myresult:
        print(myresult)

    wait = input('\n\n\nPress enter to
continue.....\n\n\n\n\n\n\n\n\n\n\n')
    main()
```

```
def report_return_books():
    a="select * from return_;"
    c=con.cursor()
    c.execute(a)
    myresult=c.fetchall()
    for i in myresult:
        print(myresult)

    wait = input('\n\n\nPress enter to
continue.....\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n')
    main()
```

```
def main():
    print("""
        LIBRARY MANAGEMENT APPLICATION
    """
```

1. ADD BOOK
 2. ISSUE OF BOOK
 3. RETURN OF BOOK
 4. DELETE BOOK
 5. DISPLAY BOOKS
 6. REPORT MENU
 7. EXIT PROGRAM
- ```
""")
```

```
choice=input("Enter Task No:.....")
print('\n\n\n\n\n\n\n\n')
if(choice=='1'):
 addbook()
elif(choice=='2'):
 issueb()
elif(choice=='3'):
 returnb()
elif(choice=='4'):
 dbook()
elif(choice=='5'):
 dispbook()
elif(choice=='6'):
 print('' R E P O R T M E N U
```

---

1. ISSUED BOOKS
  2. RETURNED BOOKS
  3. GO BACK TO MAIN MENU
- ```
\n\n\n
'''
)
```

```
choice=input("Enter Task No:.....")
print('\n\n\n\n\n\n\n\n')
if choice=='1':
    report_issued_books()
elif choice=='2':
    report_return_books()
elif choice=='3':
```

```
    main()
else:
    print("Please try again.....\n\n\n\n\n\n\n\n\n\n")
    main()
elif(choice=='7'):
    print('\n\n\n\n\n\n\n\n\n\n\n\n\n\n\nThank you and have a great day
ahead.....\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n')
else:
    print("Please try again.....\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n")
    main()

main()
```

OUTPUTS AND TABLES

➤ OUTPUTS:

1.) Add a Book:

```
LIBRARY MANAGEMENT APPLICATION
1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....1

Enter Book Name: The Immortals of Meluha
Enter Author's Name: Amish Tripathi
Enter Book Code: 101
Total Books: 10
Enter Subject: Fiction|

Book Added Successfully.....
```

```
LIBRARY MANAGEMENT APPLICATION
1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....1

Enter Book Name: The Scam
Enter Author's Name: Sucheta Dalal
Enter Book Code: 102
Total Books: 5
Enter Subject: Finanace

Book Added Successfully.....
```

2.) Issue a Book:

```
LIBRARY MANAGEMENT APPLICATION
1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....2

Enter Student Name: Ajay
Enter Reg No.: 501
Enter Book Code: 107
Enter Date: 10-01-2021

Book issued successfully to: Ajay

Press enter to continue.....
```

```
LIBRARY MANAGEMENT APPLICATION
1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....2

Enter Student Name: Akshay
Enter Reg No.: 502
Enter Book Code: 105
Enter Date: 15-01-2021

Book issued successfully to: Akshay

Press enter to continue.....
```

3.) Return of Book:

```
LIBRARY MANAGEMENT APPLICATION
1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....3

Enter Student Name: Akshay
Enter Reg No.: 502
Enter Book Code: 105
Enter Date: 19-01-2021
Book returned by: Akshay

Press enter to continue.....
```

```
LIBRARY MANAGEMENT APPLICATION
1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....3

Enter Student Name: Ajay
Enter Reg No.: 501
Enter Book Code: 107
Enter Date: 21-01-2021
Book returned by: Ajay

Press enter to continue.....
```

4.) Delete a Book:

```
LIBRARY MANAGEMENT APPLICATION

1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....4

Enter Book Code: 109
Book deleted successfully

Press enter to continue.....
```

5.) Display Books:

```
LIBRARY MANAGEMENT APPLICATION

1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....5

Book name: The Immortals of Meluha
Author: Amish Tripathi
Book code: 101
Total: 9
Subject: Fiction

Book name: The Scam
Author: Sucheta Dalal
Book code: 102
Total: 5
Subject: Finanace
```

Book name: Wings of Fire
Author: A.P.J. Abdul Kalam
Book code: 103
Total: 5
Subject: Self Reliance

Book name: One Arranged Murder
Author: Chetan Bhagat
Book code: 104
Total: 5
Subject: Thriller

Book name: Chanakya Neeti
Author: Ashwini Parashar
Book code: 106
Total: 1
Subject: Selfhelp book

Book name: Think & win like Dhoni
Author: Sfurti Sahare
Book code: 107
Total: 6
Subject: Motivation

Book name: Without Fear:Bhagat Singh
Author: Kuldeep Nayyar
Book code: 105
Total: 6
Subject: Biography

Book name: The Invisible Man
Author: H.G.Wells
Book code: 108
Total: 9
Subject: Fiction

Book name: Harry Potter series
Author: J.K.Rowling
Book code: 110
Total: 9
Subject: Drama,Mystery

Press enter to continue.....

6.) Report Menu:

```
LIBRARY MANAGEMENT APPLICATION

1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....6

REPORT MENU

1. ISSUED BOOKS
2. RETURNED BOOKS
3. GO BACK TO MAIN MENU

Enter Task No:.....
```

```
Enter Task No:.....1

('Ajay', '501', 107, '10-01-2021')
('Akshay', '502', 105, '15-01-2021')
('Ranveer', '503', 104, '19-01-2021')
('John', '504', 101, '21-01-2021')
('Abhi', '505', 108, '24-01-2021')
('Pratik', '506', 102, '28-01-2021')
('Neeraj', '507', 106, '31-01-2021')
('Rohit', '508', 104, '02-02-2021')
('Sanjay', '509', 101, '06-02-2021')
('Alia', '510', 110, '09-02-2021')
('Neha', '511', 109, '13-02-2021')
```

```
Press enter to continue.....
```

R E P O R T M E N U

1. ISSUED BOOKS
2. RETURNED BOOKS
3. GO BACK TO MAIN MENU

Enter Task No:.....2

('Akshay', '502', 105, '19-01-2021')
('Ajay', '501', 107, '21-01-2021')
('John', '504', 101, '28-01-2021')
('Ranveer', '503', 104, '31-01-2021')
('Pratik', '506', 102, '05-02-2021')

Press enter to continue.....

R E P O R T M E N U

1. ISSUED BOOKS
2. RETURNED BOOKS
3. GO BACK TO MAIN MENU

Enter Task No:.....3

L I B R A R Y M A N A G E M E N T A P P L I C A T I O N

1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

7.) Exit Program:

LIBRARY MANAGEMENT APPLICATION

1. ADD BOOK
2. ISSUE OF BOOK
3. RETURN OF BOOK
4. DELETE BOOK
5. DISPLAY BOOKS
6. REPORT MENU
7. EXIT PROGRAM

Enter Task No:.....7

Thank you and have a great day ahead.....

➤ TABLES:

1.) Select * from Books:

```
mysql> select * from books;
+----+-----+-----+-----+-----+-----+
| bname | author | bcode | total | subject |
+----+-----+-----+-----+-----+
| The Immortals of Meluha | Amish Tripathi | 101 | 9 | Fiction |
| The Scam | Sucheta Dalal | 102 | 5 | Finance |
| Wings of Fire | A.P.J. Abdul Kalam | 103 | 5 | Self Reliance |
| One Arranged Murder | Chetan Bhagat | 104 | 5 | Thriller |
| Chanakya Neeti | Ashwini Parashar | 106 | 1 | Selfhelp book |
| Think & win Like Dhoni | Sfurti Sahare | 107 | 6 | Motivation |
| Without Fear:Bhagat Singh | Kuldeep Nayyar | 105 | 6 | Biography |
| The Invisible Man | H.G.Wells | 108 | 9 | Fiction |
| Rich dad poor dad | Robert Kiyosaki | 109 | 1 | Selfhelp |
| Harry Potter series | J.K.Rowling | 110 | 9 | Drama,Mystery |
+----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

2.) Select * from Issue:

```
mysql> select * from issue;
```

name	regno	bcode	issue_date
Ajay	501	107	10-01-2021
Akshay	502	105	15-01-2021
Ranveer	503	104	19-01-2021
John	504	101	21-01-2021
Abhi	505	108	24-01-2021
Pratik	506	102	28-01-2021
Neeraj	507	106	31-01-2021
Rohit	508	104	02-02-2021
Sanjay	509	101	06-02-2021
Alia	510	110	09-02-2021
Neha	511	109	13-02-2021

11 rows in set (0.00 sec)

3.) Select * from Return_

```
mysql> select * from return_;
```

name	regno	bcode	return_date
Akshay	502	105	19-01-2021
Ajay	501	107	21-01-2021
John	504	101	28-01-2021
Ranveer	503	104	31-01-2021
Pratik	506	102	05-02-2021

```
5 rows in set (0.00 sec)
```

BIBLIOGRAPHY

❖ To develop this project many references were used:

1. INFORMATICS PRACTICES Class XII: PREETI ARORA

2. <https://www.google.com>

REMARKS

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.
